

Unit-4

X✓③

X✓①
X

⊗

X

X✓②

↗③
nearest

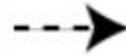
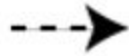
KNN ALGORITHM

↪ neighbors

✓
KNN Classifier



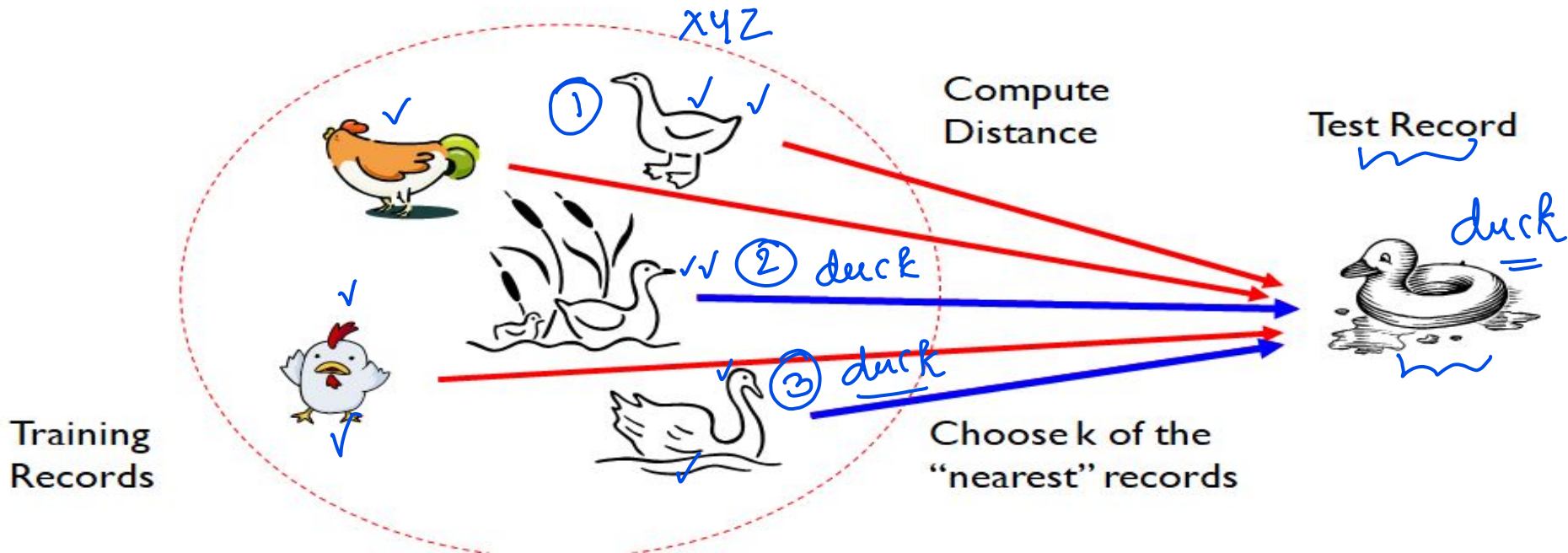
Input value



Predicted Output

Nearest Neighbor Classifiers

► Basic idea:



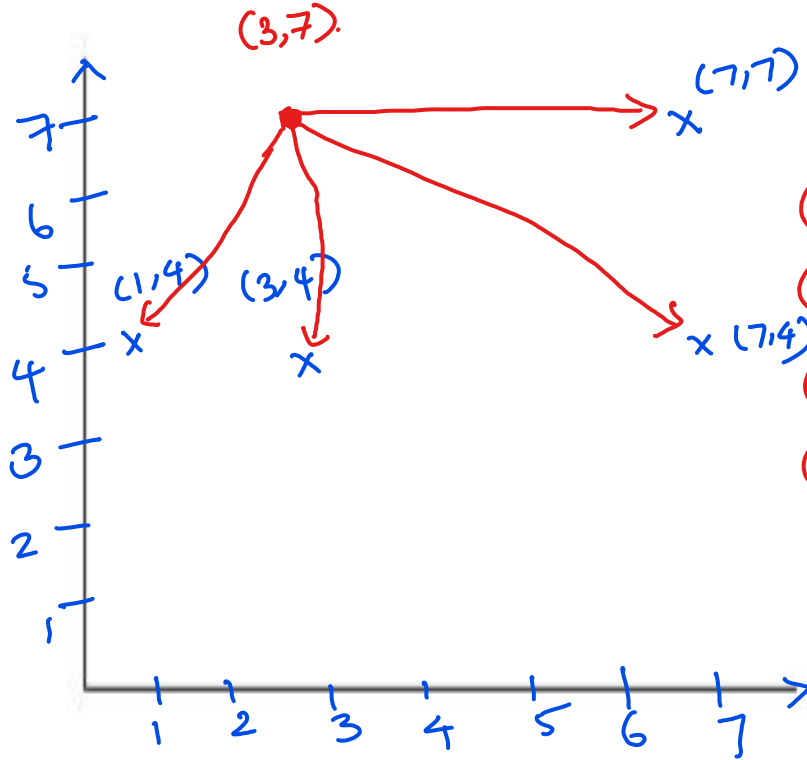
Example for Classification - Whether tissue is good? (3,7)

$k=3$

Good

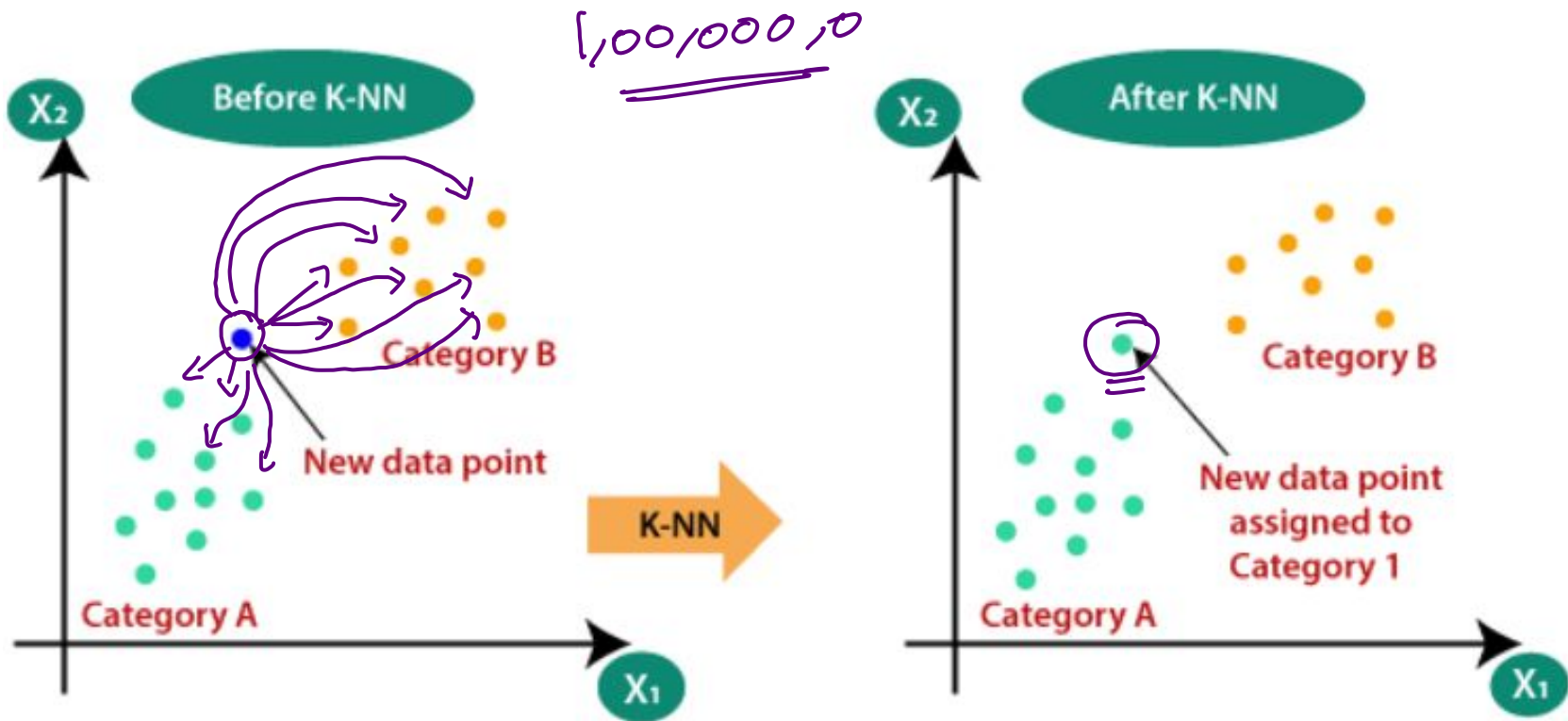
Tissue

$\checkmark$ x	$\checkmark$ y	Class $\checkmark$
$\checkmark$ 7	7	Bad
$\checkmark$ 7	4	Bad
$\checkmark$ 3	4	Good
$\checkmark$ 1	4	Good



① (7,7) $\checkmark$	$ 7-3 + 7-7 = 0$ $\checkmark$	① $\checkmark$ Bad
② (7,4)	$ 7-3 + 4-7 = 7$ $\checkmark$	Bad
③ (3,4)	$ 3-3 + 4-7 = 3$ $\checkmark$	② $\checkmark$ Good
④ (1,4)	$ 1-3 + 4-7 = 5$ $\checkmark$	② $\checkmark$ Good

0, 3, 5, 7



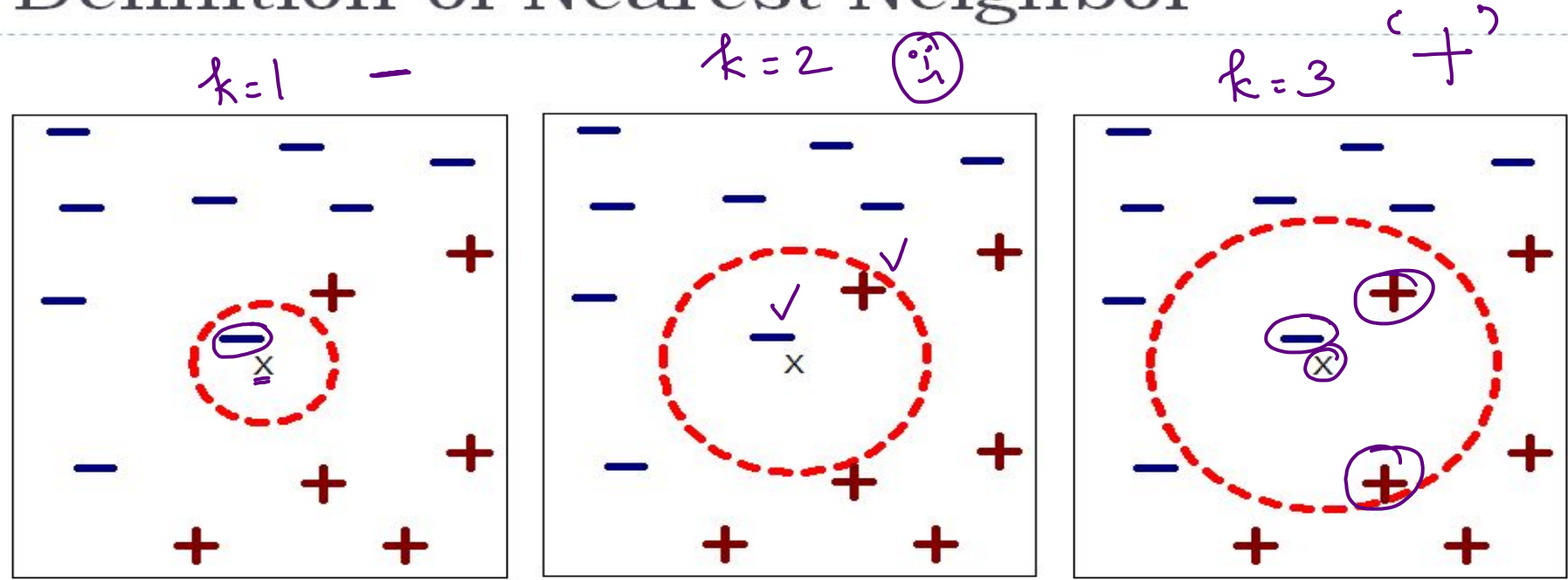
Class — majority
regression — avg -

$k=2$ → good
→ bad

Basic Idea

- k -NN classification rule is to assign to a test sample the majority category label of its k nearest training samples
- In practice, k is usually chosen to be odd, so as to avoid ties
- The $k = 1$ rule is generally called the nearest-neighbor classification rule

Definition of Nearest Neighbor



(a) 1-nearest neighbor

(b) 2-nearest neighbor

(c) 3-nearest neighbor

K-nearest neighbors of a record x are data points that have the k smallest distance to x

Steps

- **Step-1:** Select the number K of the neighbors ✓
- **Step-2:** Calculate the Euclidean distance of **K number of neighbors** ✓
- **Step-3:** Take the K nearest neighbors as per the calculated Euclidean distance. ✓
- **Step-4:** Among these k neighbors, count the number of the data points in each category. ✓
- **Step-5:** Assign the new data points to that category for which the number of the neighbor is maximum.
- **Step-6:** Our model is ready.

Key points

- K-NN is a **non-parametric algorithm**, which means it does not make any assumption on underlying data.
- It is also called a **lazy learner algorithm** because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.
- KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

Nearest-Neighbor Classifiers: Issues

- ① – The value of k , the number of nearest neighbors to retrieve
- ② – Choice of Distance Metric to compute distance between records
- ③ – Computational complexity
 - Size of training set ✓
 - Dimension of data ✓

Value of K ✓

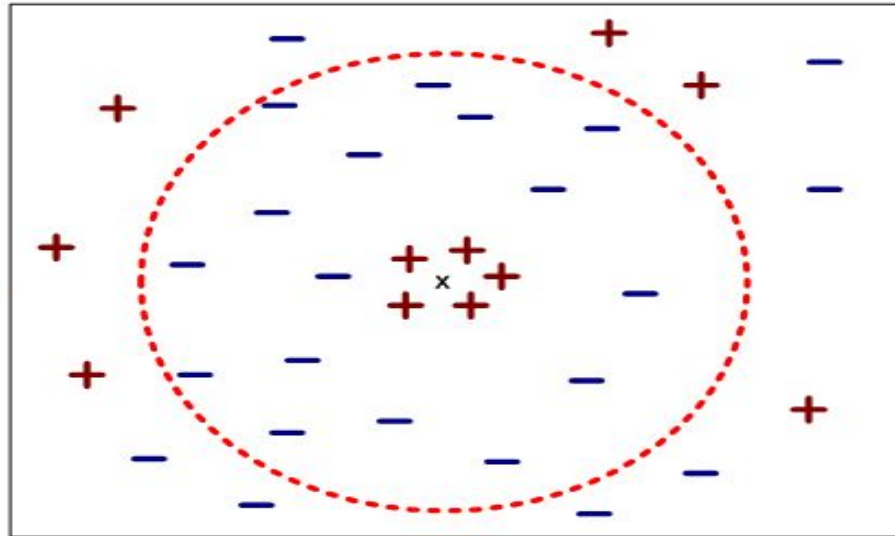
► Choosing the value of k:

- If k is too small, sensitive to noise points
- If k is too large, neighborhood may include points from other classes

Rule of thumb: ✓

$$K = \sqrt{N}$$

N: number of training points



Thanks